Network Science

Chapter 3: Random Networks

Comprehensive Study Notes — Barabási (2016)

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Contents

1 The Random Network Model (Sec. 3.2)

Core Definition

A **random network** consists of N nodes where each pair of nodes is connected with probability p. The network produced is called an **Erdős–Rényi (ER) network**, named after Paul Erdős and Alfred Rényi.

1.1 Two Equivalent Formulations

- $G(N, L)$ **model** (original Erdős–Rényi): N labeled nodes connected by exactly L randomly placed links. Average degree: $\langle k \rangle = 2L/N$.
- $G(N, p)$ **model** (Gilbert, 1959): Each pair among N nodes is connected independently with probability p.

Which model does this book use?

The $G(N, p)$ model is used throughout because real networks do not have a fixed number of links, and network properties are easier to compute analytically.

1.2 Construction Algorithm

1. Start with N isolated nodes.
2. For each of the $\frac{N(N-1)}{2}$ node pairs, draw $r \sim \text{Uniform}[0, 1]$.
3. If $r \leq p$, add a link; otherwise leave disconnected.

Key point: Two realizations with identical (N, p) look different and have different numbers of links — randomness is inherent, not a modelling artifact.

2 Number of Links and Average Degree (Sec. 3.3)

Probability of Exactly L Links (Binomial)

$$P_L = \binom{N(N-1)/2}{L} p^L (1-p)^{\frac{N(N-1)}{2} - L}$$

P_L	probability that a particular realization has exactly L links
p^L	probability that L specific pairs are connected
$(1-p)^{\cdots}$	probability that the remaining pairs are absent
$\binom{\cdots}{L}$	number of ways to arrange L links among all possible pairs

Expected Number of Links

$$\langle L \rangle = p \frac{N(N-1)}{2}$$

$\langle L \rangle$	expected (mean) number of links across realizations
p	connection probability
$\frac{N(N-1)}{2}$	total number of possible node pairs ($L_{\max}$)

Average Degree

$$\langle k \rangle = \frac{2\langle L \rangle}{N} = p(N-1)$$

$\langle k \rangle$	average degree (mean number of links per node)
p	connection probability
$N-1$	maximum possible degree of a node (can link to all others)

Implication: As p grows from 0 to 1, $\langle L \rangle$ and $\langle k \rangle$ grow linearly from 0 to their maximum values.

3 Degree Distribution (Sec. 3.4)

The **degree distribution** p_k is the probability that a randomly chosen node has exactly k links.

3.1 Exact Form: Binomial

Binomial Degree Distribution

$$p_k = \binom{N-1}{k} p^k (1-p)^{N-1-k}$$

p_k	probability a node has degree k
p^k	probability that k specific links exist
$(1-p)^{N-1-k}$	probability that the remaining $N-1-k$ potential links are absent
$\binom{N-1}{k}$	ways to choose k links from $N-1$ possible links

Moments:

$$\langle k \rangle = p(N-1), \qquad \sigma_k = \sqrt{p(1-p)(N-1)}$$

3.2 Approximation for Sparse Networks: Poisson

Most real networks satisfy $\langle k \rangle \ll N$ (sparse limit). The binomial then simplifies to:

Poisson Degree Distribution (Primary form used throughout the book)

$$p_k = \frac{e^{-\langle k \rangle} \langle k \rangle^k}{k!}$$

p_k	probability of a randomly chosen node having degree k
$e^{-\langle k \rangle}$	normalisation: ensures $\sum_k p_k = 1$
$\langle k \rangle^k$	grows with mean degree for moderate k
$k!$	factorial; grows faster than $\langle k \rangle^k$, suppressing large-k nodes
$\langle k \rangle$	**single parameter** — completely characterises the distribution

Why Poisson is preferred

- Depends on only one parameter: $\langle k \rangle$ (binomial needs both p and N).
- For fixed $\langle k \rangle$, p_k is **independent of network size** N. Networks of different sizes but the same $\langle k \rangle$ are indistinguishable.
- Simpler closed-form expressions for all moments.
- Valid when $N \gg \langle k \rangle$ (true for all large sparse networks).

Moments of the Poisson Distribution

$$\langle k \rangle = \langle k \rangle, \qquad \sigma_k = \sqrt{\langle k \rangle}, \qquad \langle k^2 \rangle = \langle k \rangle(\langle k \rangle + 1)$$

$\sigma_k = \langle k \rangle^{1/2}$ standard deviation grows as square root of mean degree

Key implication: The relative width $\sigma_k / \langle k \rangle = 1/\sqrt{\langle k \rangle}$ shrinks as $\langle k \rangle$ grows, so **all nodes have similar degrees** in a large random network.

Why Hubs Are Absent in Random Networks

Using the Stirling approximation $k! \approx \sqrt{2\pi k}\,(k/e)^k$, the Poisson formula becomes:

$$p_k \approx \frac{e^{-\langle k \rangle}}{\sqrt{2\pi k}} \left(\frac{e\langle k \rangle}{k}\right)^k$$

For $k > e\langle k \rangle$, the term $(e\langle k \rangle/k) < 1$, so p_k decreases **faster than exponentially**. High-degree hubs are virtually impossible in a random network.

Social network example ($\langle k \rangle \approx 1000$, $N \approx 7 \times 10^9$):

- Predicted $k_{\max} \approx 1185$, $k_{\min} \approx 816$, $\sigma_k \approx 31.6$
- In reality: FDR's appointment book had 22,000 contacts; Facebook users reach 5,000 friends
- **Conclusion: Random network model grossly underestimates heterogeneity.**

4 Evolution of a Random Network — Phase Transitions (Sec. 3.6)

As p (or equivalently $\langle k \rangle$) increases from 0, the network passes through four regimes separated by sharp **phase transitions**.

Giant Component

A **giant component** is a connected cluster whose size N_G scales linearly with N, i.e., $N_G/N > 0$ as $N \to \infty$. It is the unique largest connected subgraph.

Critical Condition for Giant Component Emergence (Erdős–Rényi 1959)

$$\langle k \rangle = 1 \quad \Longleftrightarrow \quad p_c = \frac{1}{N-1} \approx \frac{1}{N}$$

$\langle k \rangle = 1$	each node must have *at least one link on average* for a giant component
p_c	critical connection probability; $p_c \to 0$ as $N \to \infty$

Intuition: For a giant component, each node in it must link to at least one other. One link per node (on average) is both necessary and sufficient.

Giant Component Size (Supercritical — near critical point)

$$\frac{N_G}{N} \sim \langle k \rangle - 1, \qquad \text{equivalently} \qquad N_G \sim (p - p_c)\,N$$

N_G	size (number of nodes) of the giant component
$\langle k \rangle - 1$	how far above the critical threshold $\langle k \rangle = 1$
$p - p_c$	excess connection probability above the critical value

Full Connectivity Threshold

$$\langle k \rangle = \ln N \qquad \Longleftrightarrow \qquad p = \frac{\ln N}{N}$$

$\ln N$	average degree needed for all nodes to join the giant component
N	network size; the threshold grows slowly (logarithmically) with N

4.1 The Four Regimes

Regime	**Condition**	**Properties**
Subcritical	$0 < \langle k \rangle < 1$	Network consists of tiny isolated clusters. Largest cluster is a tree with $N_G \sim \ln N$, so $N_G/N \to 0$. Component sizes follow an *exponential* distribution — no clear "winner."
Critical Point	$\langle k \rangle = 1$	Phase transition. Largest component: $N_G \sim N^{2/3}$, still $N_G/N \to 0$. Component sizes follow a *power law*: very different sizes coexist. Analogous to a physical phase transition.
Supercritical	$1 < \langle k \rangle < \ln N$	Giant component with $N_G \propto N$ coexists with many small tree-like components. **Most real networks live here.** Near critical point: $N_G/N \sim \langle k \rangle - 1$.
Connected	$\langle k \rangle > \ln N$	All nodes absorbed into giant component; $N_G \approx N$. Network is fully connected with no isolated nodes.

5 Real Networks Are Supercritical (Sec. 3.7)

Empirical Findings

- **All** studied real networks have $\langle k \rangle > 1 \Rightarrow$ giant component always present. ✓
- **Most** have $\langle k \rangle < \ln N \Rightarrow$ should be fragmented into isolated components. Yet many (Internet, power grid) are fully connected. ×
- This is a direct **failure** of the ER model for real networks.

Network	N	L	$\langle k \rangle$	$\ln N$	**Regime**
Internet	192,244	609,066	6.34	12.17	Supercritical
Power Grid	4,941	6,594	2.67	8.51	Supercritical
Science Collab.	23,133	94,437	8.08	10.05	Supercritical
Actor Network	702,388	29,397,908	83.71	13.46	**Connected** ($\langle k \rangle > \ln N$)
Protein Interactions	2,018	2,930	2.90	7.61	Supercritical

6 Small Worlds — Average Path Length (Sec. 3.8)

Small World Phenomenon (Six Degrees of Separation)

Any two individuals on Earth can be connected through a chain of ≤ 6 acquaintances. In network language: the **average shortest path** $\langle d \rangle$ between two randomly chosen nodes scales only **logarithmically** with N — much shorter than in regular lattices.

Average Path Length Formula

$$\langle d \rangle \approx \frac{\ln N}{\ln\langle k \rangle}$$

$\langle d \rangle$	average shortest path (number of hops) between any two random nodes
N	total number of nodes; $\ln N$ grows very slowly with N
$\langle k \rangle$	average degree; denser networks ($\uparrow \langle k \rangle$) have shorter paths ($\downarrow \langle d \rangle$)
$\ln\langle k \rangle$	acts as a divisor: higher connectivity compresses distances

6.1 Derivation Intuition

From any starting node, the number of nodes reachable within d steps grows geometrically:

$$N(d) \approx \langle k \rangle^d$$

Setting $N(d_{\max}) = N$ and solving: $\langle k \rangle^{d_{\max}} = N \Rightarrow d_{\max} = \ln N / \ln\langle k \rangle$.

Social Network Calculation

Using $N \approx 7 \times 10^9$ and $\langle k \rangle \approx 10^3$:

$$\langle d \rangle \approx \frac{\ln(7 \times 10^9)}{\ln(10^3)} = \frac{22.78}{6.91} \approx 3.3$$

All humans are within **3–4 handshakes**. Milgram's 1967 experiment found median = 5.2. Facebook 2011 study (721M users): $\langle d \rangle \approx 4.74$ ("four degrees of separation").

6.2 Why Is This Surprising? Comparison to Lattices

Network type	**Scaling of $\langle d \rangle$**	**Note**
Random network	$\langle d \rangle \sim \ln N$	Logarithmic — **small world**
1D lattice	$\langle d \rangle \sim N$	Linear — extremely large
2D lattice (grid)	$\langle d \rangle \sim N^{1/2}$	Polynomial
3D lattice (cube)	$\langle d \rangle \sim N^{1/3}$	Polynomial
d-dim. lattice	$\langle d \rangle \sim N^{1/d}$	General form

For the social network as a 2D lattice: $\langle d \rangle \approx (7 \times 10^9)^{1/2} \approx 83,666$ — orders of magnitude larger than the ≈ 3 predicted by random network theory.

WWW: 19 Degrees of Separation

Measurements show $\langle d \rangle \approx 0.35 + 0.89 \ln N$ for the Web. With $N \approx 8 \times 10^8$ (1999) this gives $\langle d \rangle \approx 18.69$ — the "19 degrees of separation." For $N \sim 10^{12}$ (current estimate), $\langle d \rangle \approx 25$. The Web has a larger $\langle d \rangle$ than the social network because it has smaller $\langle k \rangle$ and larger N.

7 Clustering Coefficient (Sec. 3.9)

Local Clustering Coefficient

For node i with k_i neighbours:

$$C_i = \frac{2L_i}{k_i(k_i-1)}$$

L_i	actual number of links among i's k_i neighbours
$k_i(k_i-1)/2$	maximum possible links among k_i neighbours
$C_i = 0$	none of i's neighbours know each other
$C_i = 1$	all of i's neighbours are fully interconnected

Clustering Coefficient in a Random Network

In a random network, the probability that any two of i's neighbours are connected is just p. Expected number of links among neighbours: $\langle L_i \rangle = p\, k_i(k_i-1)/2$.

$$C_i = p = \frac{\langle k \rangle}{N}$$

C_i	equals p for *every* node (independent of k_i!)
$\langle k \rangle$	average degree
N	network size; C_i decays as $1/N$ for fixed $\langle k \rangle$

★ ★ Random Vs Real

Two Predictions That Fail for Real Networks

1. **Prediction 1**: $C \propto 1/N$ for fixed $\langle k \rangle$ — larger networks have smaller C.
   Reality: $\langle C \rangle$ of real networks is essentially *independent* of N.
2. **Prediction 2**: C is the same for all nodes regardless of their degree.
   Reality: $C(k)$ *decreases* with k in real networks — hubs have lower clustering than low-degree nodes.

Real networks have **far higher clustering** than an ER random network of the same N and L.

8 The Watts–Strogatz (Small World) Model (Box 3.9)

Proposed to explain the coexistence of:

- Small world property: $\langle d \rangle \sim \ln N$ (present in random networks)
- High clustering: $\langle C \rangle \gg \langle k \rangle / N$ (absent in random networks) → present in regular lattice

WS Construction Algorithm

1. Create a ring of N nodes, each connected to its k nearest neighbours. ($\langle C \rangle = 3/4$, large $\langle d \rangle$)
2. With probability p, rewire each link to a randomly chosen node.
3. $p = 0$: regular lattice (high C, large $\langle d \rangle$)
4. $p = 1$: random network (low C, small $\langle d \rangle$)
5. $0.001 < p < 0.1$: random long-range links sharply reduce $\langle d \rangle$ while C remains high. **Both properties coexist.**

Limitations of WS Model

- Predicts a Poisson-like, bounded degree distribution — no hubs.
- Predicts C independent of node degree k — contradicts real networks where $C(k)$

decreases.
- The correct degree distribution must be addressed before $C(k)$ can be explained (addressed in later chapters).

9 Real Networks vs. Random Networks: Full Summary (Sec. 3.10)

Property	**ER Prediction**	**Real Networks**
Degree distribution	Poisson: narrow peak at $\langle k\rangle$, no hubs, p_k drops faster than exponentially for large k	Heavy-tailed: hubs present, many low-degree nodes, power-law-like tails
Giant component	Exists iff $\langle k\rangle > 1$	Always present ($\langle k\rangle \gg 1$) ✓
Fragmentation	Multiple components unless $\langle k\rangle > \ln N$	Often connected even when $\langle k\rangle < \ln N$ ×
Average path length	$\langle d\rangle \approx \ln N/\ln\langle k\rangle$	Reasonably well approximated ✓
Clustering	$C = \langle k\rangle/N$, decreases as $1/N$, C independent of node degree k	C large, independent of N; $C(k)$ decreases with k ×

Core Conclusion

The random network model correctly explains **only the small-world phenomenon**. It fails on degree distribution, clustering coefficient, and full connectivity. **No known real network is accurately described by the ER model.**

Why study it then? The ER model is the essential *null model*. If a property is present in random networks, randomness can explain it. If absent, the property signals deeper structural order requiring explanation in more sophisticated models.

10 Binomial Distribution Reference (Box 3.3)

Binomial Distribution

For N independent trials, each with success probability p:

$$P(X = x) = \binom{N}{x} p^x (1-p)^{N-x}$$

$$\langle x\rangle = Np, \qquad \langle x^2\rangle = p(1-p)N + p^2N^2, \qquad \sigma_x = \sqrt{Np(1-p)}$$

$\langle x\rangle$	mean (first moment)
$\langle x^2\rangle$	second moment
σ_x	standard deviation (spread of the distribution)

These results directly underpin $\langle L\rangle$, $\langle k\rangle$, and σ_k for random networks.

11 All Key Formulas at a Glance

Quantity	**Formula**	**Section**
Max. possible pairs	$L_{\max} = \dfrac{N(N-1)}{2}$	3.3
Expected links	$\langle L \rangle = p\,\dfrac{N(N-1)}{2}$	3.3
Average degree	$\langle k \rangle = p(N-1)$	3.3
Degree dist. (binomial)	$p_k = \binom{N-1}{k} p^k (1-p)^{N-1-k}$	3.4
Degree dist. (Poisson)	$p_k = \dfrac{e^{-\langle k \rangle} \langle k \rangle^k}{k!}$	3.4
Degree std. deviation	$\sigma_k = \sqrt{\langle k \rangle}$	3.5
Critical prob.	$p_c = \dfrac{1}{N-1} \approx \dfrac{1}{N}$	3.6
Giant component cond.	$\langle k \rangle > 1$	3.6
GC size (supercritical)	$\dfrac{N_G}{N} \sim \langle k \rangle - 1$	3.6
Full connectivity	$\langle k \rangle > \ln N$	3.6
Nodes within d hops	$N(d) \approx \langle k \rangle^d$	3.8
Average path length	$\langle d \rangle \approx \dfrac{\ln N}{\ln \langle k \rangle}$	3.8
Clustering coefficient	$C_i = p = \dfrac{\langle k \rangle}{N}$	3.9

12 Historical Notes

- **Anatol Rapoport (1951)**: First to study random networks mathematically; observed the abrupt subcritical-to-supercritical transition.
- **Erdős & Rényi (1959–1968)**: Eight landmark papers merging probability theory, combinatorics, and graph theory — founded random graph theory.
- **Edgar Gilbert (1959)**: Independently introduced the $G(N, p)$ model the same year.
- **Kochen & de Sola Pool (unpublished, circulated 1950s)**: First mathematical formulation of the small-world problem and its sociological implications.
- **Stanley Milgram (1967)**: Empirical small-world experiment; median path length = 5.2.
- **Watts & Strogatz (1998)**: Extended ER to explain coexistence of high clustering and

small-world property.

- **Facebook study (2011)**: 721M users, $\langle d \rangle \approx 4.74$ — "four degrees."
